

$^{40}\text{Ca}(\mu^-, \nu\alpha 2n\gamma)$ 2006Me08

2006Me08: Muons were generated from the decay of 90-MeV/c π^- beam in a 6-m, 1.2-T superconducting solenoid and were selected by a bending magnet at the M9B beamline at TRIUMF. The target was 15-g natural calcium. γ rays and x rays were detected using a 186-cm³ HPGe detector. Charged particles caused by electrons from muon decay were tagged by a plastic scintillator in front of the HPGe detector. Measured E_γ , I_γ , $E(x \text{ ray})$, $I(x \text{ ray})$. Deduced levels, muon capture yields.

Muonic Lyman series for natural Calcium

μ x ray	Energy	Intensity in percent
2p-1s	783.659 25	83.8 10
3p-1s	940.63 10	6.2 2
4p-1s	995.48 10	2.0 1
5p-1s	1020.81 10	2.0 1
6p-1s	1034.62 10	1.8 1
7p-1s	1042.71 20	1.4 1
(8- ∞)p-1s	1046-1063	2.8 4

Muonic Balmer series for natural Calcium

μ x ray	Energy	Intensity in percent
3d-2p	157.35 13	64.5 9
4d-2p	212.03 10	8.85 20
5d-2p	237.31 10	4.34 20
6d-2p	251.06 10	3.29 20
7d-2p	259.45 10	1.37 20
(8- ∞)d-2p	261-277	1.4 3

 ^{34}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>
0	0^+
146.4	3^+
461.0	1^+
665.6	1^+
1230.3	2^+

† From Adopted Levels. Level energies are rounded-off values.

 $\gamma(^{34}\text{Cl})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	Comments
461.0	0.10 5	461.0	1^+	0	0^+	
564.7	0.08 8	1230.3	2^+	665.6	1^+	
665.5	<0.1	665.6	1^+	0	0^+	
769.3		1230.3	2^+	461.0	1^+	I_γ : intensity not listed in 2006Me08; the peak is obscured by other transitions.
1083.9	<0.15	1230.3	2^+	146.4	3^+	

† Rounded-off values from Adopted Gammas.

‡ Percent γ -ray yield per muon capture (2006Me08).

$^{40}\text{Ca}(\mu^-, \nu\alpha 2n\gamma)$ 2006Me08

Level Scheme

Intensities: percent γ -ray yield per muon capture

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
—→ $I_\gamma < 10\% \times I_\gamma^{\max}$
—→ $I_\gamma > 10\% \times I_\gamma^{\max}$

